

AMENDMENT NO. 1 APRIL 1989
TO
IS : 6921 - 1973 METHODS OF SAMPLING AND
TEST FOR LAC AND LAC PRODUCTS

(*Page 16, clause 7*) — Insert the following after 7:

7.0 General — The colour index of seed lac and shellac is determined by the method as described in 7.1 and colour of index shellac only is determined by the method as described in 7.2 which may be used as referee method for determination of colour index of shellac.

7.1 Method

(*Pages 16 to 19 clause 7.0 to 7.3.3*) — Renumber the clauses **7.0, 7.1, 7.1.1, 7.1.2, 7.1.3, 7.2, 7.2.1, 7.2.1.1, 7.2.1.2, 7.2.2, 7.3, 7.3.1, 7.3.2,** and **7.3.3** as **7.1.0, 7.1.1, 7.1.1.1, 7.1.1.2, 7.1.1.3, 7.1.2, 7.1.2.1, (a), (b), 7.1.2.2, 7.1.3, 7.1.3.1, 7.1.3.2,** and **7.1.3.3** respectively and delete clause number **7.2.2.1**.

(*Page 19, clause 7.3.3*) — Insert the following after **7.3.3**:

7.2 Method 2

7.2.0 Outline of the Method — This method consists of the measurement optical density of an alcoholic shellac solution (cone 10 g/l) at a particular wavelength in the visible range, which after multiplication by 136.9 gives the value of colour index.

7.2.1 Apparatus

7.2.1.1 Spectrophotometer — Any spectrophotometer/Colorimeter (Grating type) in the visible range (400 — 700 nm).

7.2.1.2 Volumetric flasks — with stopper (ground joint) 100 ml.

7.2.1.3 Fipette — 1 ml.

7.2.2 Reagents

7.2.2.1 Alcohol — Ethyl alcohol (absolute) or 95 percent (by volume) rectified spirit (conforming to IS : 323 - 1959*) or 95 percent (by volume) denatured spirit (conforming to IS : 324-1959†) provided that it is colourless.

7.2.3 Procedure

7.2.3.1 Preparation of solution — Weigh accurately 1 g of the prepared

*Specification for rectified spirit (*revised*).

†Specification for ordinary denatured spirit (*revised*).

sample (*see* 3.3.2) of shellac and transfer the material to the 100 ml volumetric flask. Add 60 - 70 ml of alcohol and shake the flask vigorously as soon as the alcohol is added till the shellac is completely dissolved. Add more solvent, and finally make up the volume up to the mark of the volumetric flask. Filter the solution in an ordinary funnel using medium grade filter paper (preferably Whatman No. 1) and keeping the funnel covered (best result is obtained if the filtration is done under saturated vapour pressure of the solvent). Discard the first 15 ml of the clear filtrate.

7.2.3.2 Transfer one ml of the filtrate by means of a pipette to the 10 ml volumetric flask. Add alcohol to it and make up the volume up to the mark of the flask.

7.2.3.3 *Measurement of optical density* — Switch on the spectrophotometer/colorimeter. After the warming up period of the instrument, set the wavelength at 425 nm, match the cuvettes with the alcohol used for the preparation of the solution. Transfer a portion of the above diluted solution to one of the cuvettes. Note down the value of the optical density.

7.2.4 *Calculation*

Colour index = optical density $\times$ 136.9